



RUDRAEDGE AEROSPACE

Project - Rudra Edge RE-T1(Spectra) Daily Engineering Log

1. TEAM MEMBER INFORMATION

Engineer Name: Suraj Nadda

Designation / Role: Software & Data Engineer

Task Order Reference: 10 October 2026 – Telemetry scheme design

2. DAILY LOG ENTRY FORM

- **Date:** 10 October 2026

- **1. Assigned Task:**

Teklemetry schema design(timestamps, sensors, IDs, status masks)

- **2. What Was Studied:**

Studied the purpose of a telemetry schema and how it maintains a consistent structure for sensor records. Reviewed JSON format, required fields, data types, timestamps, sensor IDs, and measurement fields such as temperature, pressure, altitude, and speed. Also studied how status masks can represent record-validation conditions through individual bits.

- **3. What Was Implemented:**

Created telemetry_schema.json to define the telemetry fields and status-mask meanings, along with sample_telemetry.json as an example record. Developed a Python validator to check required fields, data types, timestamp format, and status-mask range. Tested the validator using a valid sample and an intentionally incomplete sample; the valid record passed, while the incomplete record was rejected because sensor_id was missing.

- **4. Tools / Libraries Used:**

VS Code, PowerShell terminal, Python 3.14.6, JSON, json, pathlib

- **5. Output File Name-** suraj_rudra_task/Day6/code/validate_schema.py

Path-https://github.com/surajnadda29/suraj_rudra_t1.git

- **6. Problems Encountered:**

During testing, the incomplete telemetry record was rejected because the required sensor_id field was missing. This showed that a record may contain other telemetry values but still fail validation when a mandatory identifier is absent.

- **7. How Problem Was Resolved:**

Checked the validator's rejection message and confirmed that sensor_id was listed as a required field. Kept the incomplete record as a negative test case to verify that missing mandatory fields are detected correctly. The complete sample was then tested separately and passed the configured validation checks.

- **8. Next Task Scheduled:**

(11 October 2026) – Synthetic UAV flight telemetry generation with realistic noise models

- **9. Personal Expenditure: ₹0 (Verified Mandate)**

3. ENGINEER SELF-DECLARATION & SIGN-OFF

I hereby formally declare that today's technical progression, synthetic dataset code, and local repository paths have been documented truthfully in compliance with Section 6 (Common Technical Standards). I certify with absolute transparency that my personal financial expenditure for today's assigned task is exactly ₹0, utilizing strictly local open-source infrastructure.

Candidate/Engineer Signature:  _____
Full Name: Suraj Nadda

Date: 10/10/2026

PROJECT LEAD VERIFICATION BOX — TO BE FILLED BY SARVESH SHARMA ONLY

I have reviewed today's technical output, local repository commits, and verified the project compliance standards for this day's assigned task.

Task Evaluation Status:

- ☐ Approved / Completed
- ☐ Needs Revision / Delayed
- ☐ Technical Block Encountered

Operational Expenditure Confirmed: ₹0 (Strict Zero-Cost Compliant)

Remarks/Notes: _____

Project Lead Signature: _____ Date: ____/____/2026